Chapter test

8 Gas exchange in humans

Name: _____ ( ) Class: _____ Date: _____

Total marks: 40 marks
Time allowed: 40 minutes
Answer ALL questions.

Score: / 40

A. Concept checking (10 marks, 1 mark each)

1. Cigarette smoke contains chemicals which could damage the cilia on the surface of the respiratory tract. Which of the following consequences may be resulted?

   A. The air exhaled by the smoker will be warmer.
   B. The air exhaled by the smoker will contain less oxygen.
   C. The air exhaled by the smoker will contain less carbon dioxide.
   D. The smoker will suffer from respiratory infections more easily. ☐

2. Which of the following statements about haemoglobin is ***incorrect***?

   A. It is an iron-containing pigment.
   B. It is made up of polypeptide chains.
   C. It binds with oxygen under high oxygen concentrations.
   D. It has a biconcave disc shape. ☐

3. The table shows the composition of inhaled air and exhaled air.

	Inhaled air	**Exhaled air**
Nitrogen	P	79%
Oxygen	21%	Q
Carbon dioxide	0.04%	R

Which of the following correctly shows the values of P, Q and R?

	P	***Q***	***R***
A.	78%	16%	18%
B.	78%	16%	4%
C.	78%	4%	4%
D.	62%	0%	18%

☐

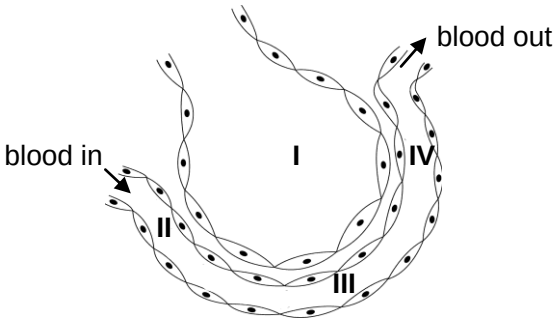
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★ 4. Which of the following is/are adaptive feature(s) of the human breathing system to trap dust particles in the inhaled air?

(1) Hairs are present in the nostrils.
(2) The wall of nasal cavity has cilia.
(3) The inner surfaces of the air sacs are covered with mucus.

A. (1) only
B. (1) and (2) only
C. (1) and (3) only
D. (2) and (3) only ☐

Directions: Questions 5 and 6 refers to the diagram below, which shows an air sac and the capillary surrounding it.



★ 5. The highest oxygen concentration can be found at

A. I.
B. II.
C. III.
D. IV. ☐

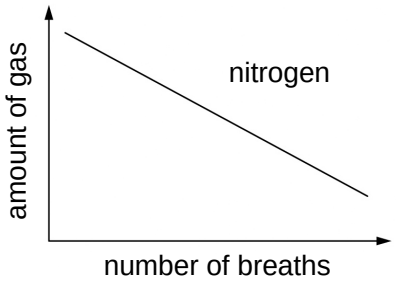
★ 6. Which of the following features help maintain the oxygen concentration gradient between the air in the air sac and the blood in the capillary?

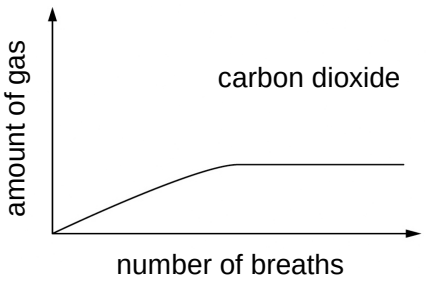
(1) The air sac wall is one-cell thick.
(2) Blood flows continuously in the capillary.
(3) Ventilation helps move air into and out of the air sac.

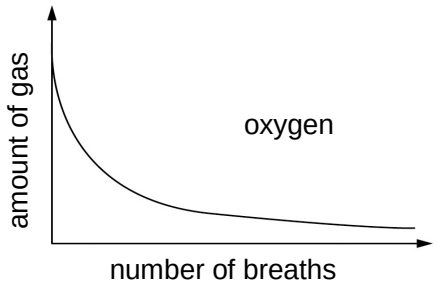
A. (1) and (2) only
B. (1) and (3) only
C. (2) and (3) only
D. (1), (2) and (3) ☐

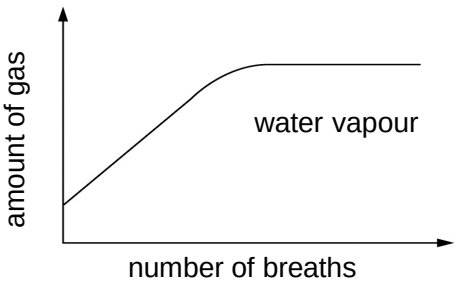
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★ 7. Mary is breathing into and out of an airtight bag. Which of the following graphs about the change in the respective gas component inside the bag is ***incorrect***?

A.


B.


C.


D.


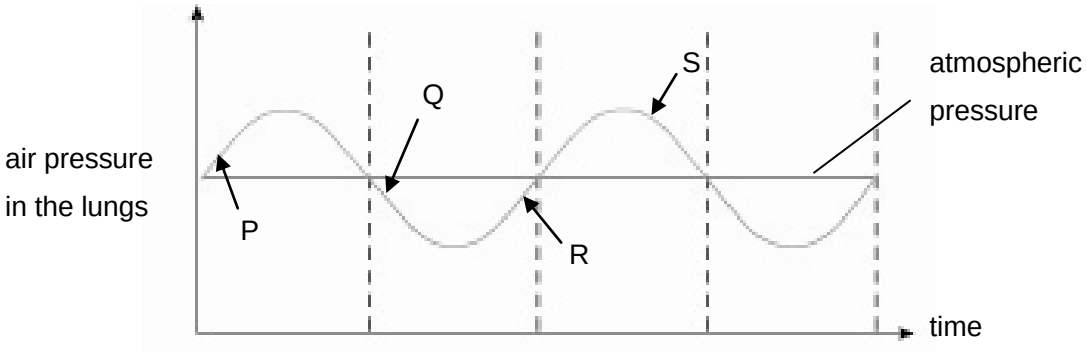
( )

★★ 8. For an oxygen molecule to move from an air sac into the haemoglobin molecules inside a red blood cell, how many times does it need to cross the cell membrane?

A. 2 times
B. 3 times
C. 4 times
D. 5 times

( )

Directions: Questions 9 and 10 refers to the graph below, which shows the changes in the air pressure in the lungs during two complete cycle of breathing.



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★★ 9. The volume of the lung is the largest at

A. P.
B. Q.
C. R.
D. S. ☐

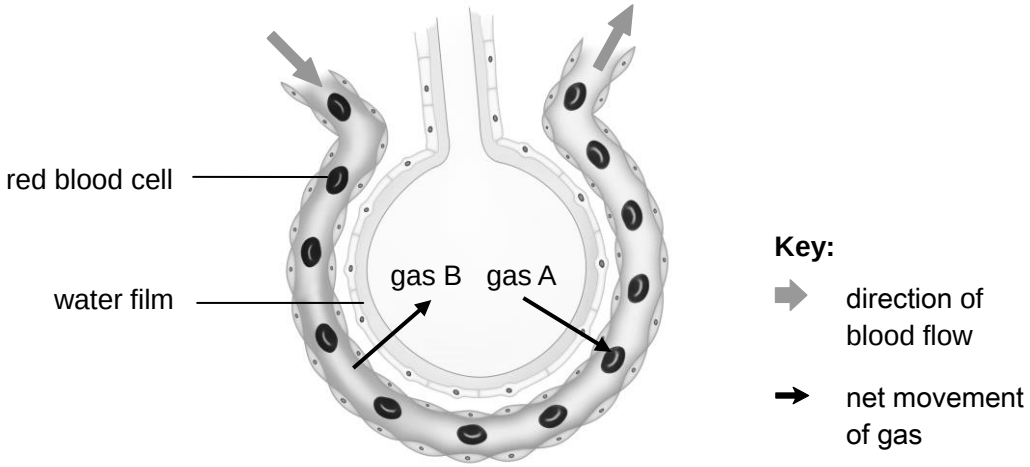
★★ 10. At which point do the ribs start to be lowered?

A. P
B. Q
C. R
D. S ☐

Question	1	2	3	4	5	6	7	8	9	10
Related section	8.1	8.3	8.3	8.1	8.2	8.2	8.3	8.2	8.4	8.4

B. Problem solving (21 marks)

11. The diagram below shows a structure found in the human lungs.



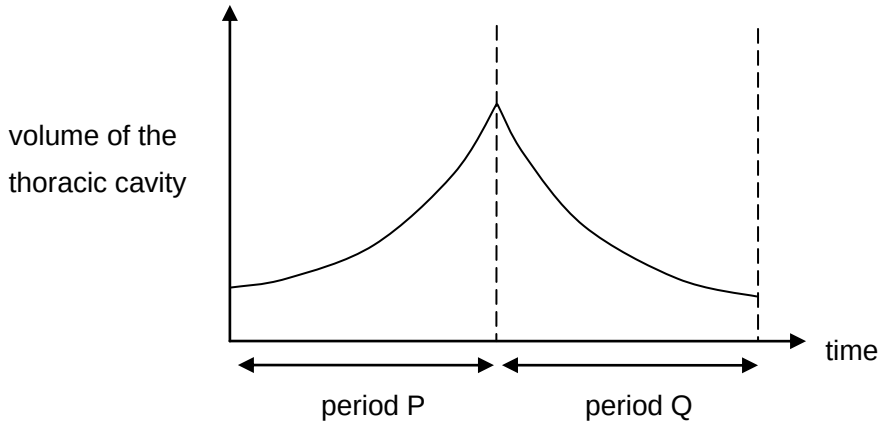
(a) Name gases A and B. (2 marks)

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(b) Describe how gas A is transported in the blood to different parts of the body. (3 marks)

(Total: 5 marks)

★ 12. The graph below shows the change in the volume of the thoracic cavity of a healthy person in one single breath.



(a) During which time period, P or Q, did inhalation occur? (1 mark)

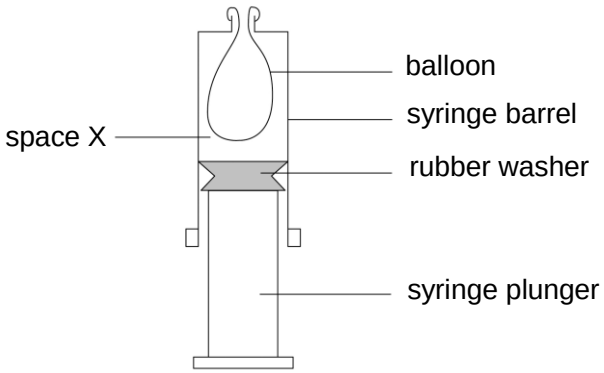
(b) Explain how the change in the thoracic cavity during period P is brought about and how the change leads to ventilation. (4 marks)

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(c) During an asthma attack, the lumen of the bronchi becomes narrower and the patient will have difficulty in breathing. Describe and explain how the changes in the volume of the thoracic cavity of an asthma patient would be different from those shown in the graph. (4 marks)

(Total: 9 marks)

★★13. The diagram below shows a model used to demonstrate how air enters the lungs during breathing. When the syringe plunger is pulled downwards, the balloon swells up with air.



(a) Which part of the human body does space X represent? (1 mark)

(b) The model cannot truly represent the breathing action of humans.

(i) Give ***two*** reasons to explain why the rubber washer cannot truly represent the diaphragm in breathing movements. (4 marks)

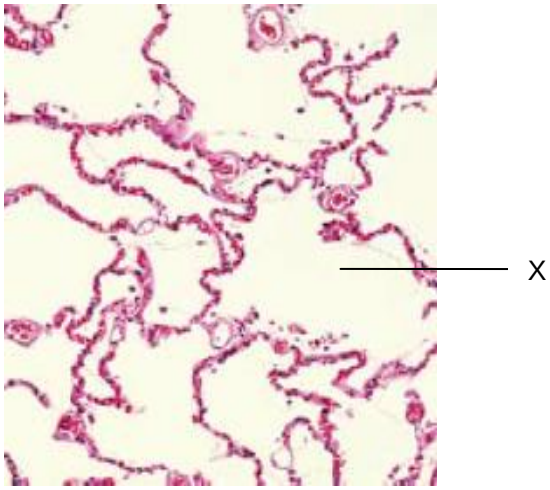
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(ii) Give ***one*** reason to explain why the syringe barrel is a poor representation of the thoracic wall. (2 marks)

(Total: 7 marks)

C. Exam challenge (9 marks)

★ 14. The photomicrograph below shows a section of a human lung.



(a) Construct a flowchart to show the path of air passing from the atmosphere to structure X of the lung. (2 marks)

(b) With reference to ***two*** observable features in the photomicrograph, explain how X is structurally adapted to the exchange of gases. (4 marks)

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(c) Patients with severe acute respiratory syndrome (SARS) have their lung tissues filled with fluid. Explain how this affects the blood oxygen content of the SARS patients. (3 marks)

(Total: 9 marks)

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Marking scheme

A. Concept checking (10 marks)

1.	D	2.	D	3.	B	4.	A	5.	A
6.	C	7.	A	8.	D	9.	C	10.	A

(1 mark each)

B. Problem solving (21 marks)

11. (a) Gas A: oxygen (1)

Gas B: carbon dioxide (1)

(b) Gas A (oxygen) diffuses into red blood cells. (1)

It binds with haemoglobin to form oxyhaemoglobin. (1)

The oxygenated blood is transported to the heart (through the pulmonary veins) and is pumped by heart to different parts of the body. (1)

(Total: 5 marks)

12. (a) period P (1)

(b) Intercostal muscles contract; the ribs move upwards and outwards. (1)

Diaphragm muscles contract; the diaphragm becomes flattened. (1)

The volume of the thoracic cavity increases and the lungs expand. (1)

The air pressure in the lungs becomes lower than atmospheric pressure. (1)

Air therefore rushes into the lungs.

(c) Less air can enter the lungs in each breathe since the lumen of the bronchi becomes narrower. (1)

The patient needs to breathe faster and deeper in order to obtain enough oxygen and remove carbon dioxide. (1)

Therefore, the duration of each breath would become shorter, and (1)

the depth of breathing / the change in the volume of the thoracic cavity would become larger. (1)

(Total: 9 marks)

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13. (a) thoracic cavity / pleural cavity (1)

(b) (i) The movement of the rubber washer is controlled by hand (1)

while the movement of the diaphragm is automatic. (1)

The rubber washer is moved downwards (1)

while the diaphragm flattens during inhalation. (1)

(ii) The syringe barrel is rigid / immovable (1)

while the thoracic wall can move during breathing. (1)

(Total: 7 marks)

C. Exam challenge (9 marks)

14. (a) atmosphere → nasal cavity → pharynx → (1)

trachea → bronchus → bronchiole → structure X (air sac) (1)

(b) The wall of structure X is thin (one-cell thick). (1)

This provides a short distance for diffusion of gases. (1)

Capillaries are present around structure X. (1)

The rich supply of blood transports gases to and away from X readily to maintain steep concentration gradients for diffusion of gases. (1)

(c) The fluid accumulated in the lung tissues reduces the surface area of the air sacs. / The fluid increases the diffusion distance of oxygen. (1)

The rate of oxygen diffusion across the air sacs decreases.. (1)

Hence, the blood oxygen content decreases. (1)

(Total: 9 marks)

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